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Welcome to another installment of CodeSport. This month's column focuses on computational complexity and the lower bounds for algorithms. In particular, we'll show that any algorithm to find the maximum in an array of N elements has a lower bound of $O(N)$ by using an adversary argument.

Thanks to all the readers who sent in their comments on the problems we discussed in last month's column, in which the takeaway question was a variant of the 3-sum problem. You were given three sets of numbers A , B and C , each containing N numbers. Readers were asked to come up with an algorithm to determine whether there was a triple $a \in A$, $b \in B$ and $c \in C$ such that $a + b = c$? It is quite easy to come up with an $O(N^2 \log N)$ algorithm. But you need to come up with an $O(N^2)$ algorithm for this.

Instead of a separate algorithm for our takeaway problem, we will first look at how we can reduce the 3-sum problem to this. Given a set S of n integers, do there exist three elements $\{a, b, c\} \in S$ such that $a + b + c = 0$? Let us denote the takeaway problem as PR2 and the 3-sum problem as PR1. Since we know that the 3-sum problem has a lower bound of $O(N^2)$, PR2 also is equally hard. Hence, the best we can hope for to solve PR2 is an $O(N^2)$ algorithm. Recall that the reduction from a 3-sum problem to PR2 can be given by $A=S$, $B=S$ and $C=-S$. Hence, $a \in A$, $b \in B$, $c \in C$, and a solution of PR2 which satisfies $a + b = c$, will also satisfy PR1. Now, we would like to look at how we can reduce PR1 to PR2; i.e., given a problem instance of PR1, we want to formulate a problem instance of PR2 that can be reduced to PR1. In that case, we can use the solution of the 3-sum problem to solve PR2.

We create one set S such that whenever three elements in S add up to zero, there are three elements $a \in A$, $b \in B$ and $c \in C$, such that $a+b = c$. For the sake of simplicity, let us assume that all elements are positive. We

define a value m , which is equal to twice the maximum element among all members of A , B and C . We now try and construct set S from the elements of A , B and C as follows: For each element $a \in A$, we add an element $a' = a + m$ to S . For each element $b \in B$, we add an element $b' = b$ to S . And for each element $c \in C$, we add an element $c' = (-c - m)$ to S . We can see that if $(a+b+c)$ is equal to zero, then $(a' + b' + c')$ is also equal to zero. In order to construct a solution of PR2 from PR1, we need to show that whenever there are three elements in S that add up to zero, then the three elements came from each of the three different sets A , B and C .

Based on the way we have created a matching element in S for each element in A , B and C , we can see that a' lies between $0.5m$ to $1.5m$; b' lies between 0 to $0.5m$; and c' lies between $-1.5m$ to $-0.5m$. Now consider three elements x, y, z in S such that $x + y + z = 0$. Now, at the most, one of the elements in (x, y, z) could have been constructed from an element in A since, otherwise, the sum would be at least $2m - 1.5m = 0.5m$. Similarly, only one element could have been constructed from C , since otherwise the sum would be $-2m + 1.5m = -0.5m$. Also note that, out of (x, y, z) , one element must come from C , since elements constructed from A and B are always positive and we need at least one negative element to make the sum zero. This negative element can only come from C . Now, if we consider the possibility that one element in (x, y, z) is constructed from C and the other two are constructed from B , then again the sum of the two positive elements is at most m , whereas the element constructed from C is smaller than m , so the sum cannot